COVID-19 Global Impact Dashboard

Project Report

This report summarizes the project objectives, dataset, preprocessing steps, exploratory analysis, dashboard, findings, conclusion, and possible future enhancements.

1. Introduction

The COVID-19 Global Impact Dashboard project analyzes reported COVID-19 cases and deaths across different locations using data analysis and visualization techniques. The project uses the Our World in Data COVID-19 dataset to explore global trends and the impact of the pandemic in India.

2. Objectives

- Analyze COVID-19 cases and deaths across different locations.
- Study daily COVID-19 case and death trends in India.
- Calculate and examine the case fatality rate.
- Visualize findings using Python and Power BI.
- Present data-driven insights through a dashboard.

3. Dataset Description

The dataset was obtained from Our World in Data. It contains information such as location, date, total cases, new cases, total deaths, new deaths, and population-related indicators.

4. Tools and Technologies

- Python
- Google Colab
- Pandas and NumPy
- Matplotlib and Seaborn
- Microsoft Power BI
- CSV dataset

5. Data Preprocessing

The dataset was copied for cleaning, the date column was converted to datetime format, duplicate records were removed, rows with missing dates or locations were removed, and records were sorted by location and date. Missing values in important columns were examined.

6. Exploratory Data Analysis

The analysis included comparisons of total COVID-19 cases and deaths across locations, daily new cases and deaths in India, total cases per million people, and the case fatality rate in India.

7. Dashboard

A Power BI dashboard was developed to present key performance indicators and visualizations of COVID-19 data. The dashboard includes cards for cases, deaths, locations, and the latest data date, along with charts for comparing locations and exploring case trends.

8. Key Findings

- The visualizations show differences in reported COVID-19 cases and deaths across locations.
- India's daily case and death trends show major peaks around 2021.
- The case fatality rate generally declined over the period shown in the analysis.

9. Conclusion

This project demonstrates how data preprocessing, exploratory data analysis, and visualization can turn raw data into meaningful insights. Python and Power BI were used to explore COVID-19 trends and communicate findings through charts and a dashboard.

10. Future Enhancements

The project could be enhanced with additional interactive filters, more detailed regional comparisons, and regularly updated data.